

1. Techniques to Reduce Parasitic Capacitance

a. Minimize Wire Length and Width

- i. Shorter interconnects reduce resistance (R) and capacitance (C), lowering RC delay.
- ii. Narrower metal wires decrease parasitic capacitance to the substrate.

b. Use Higher Metal Layers

- i. Routing signals on higher metal layers reduces coupling capacitance and substrate capacitance.
- ii. This minimizes the RC time constant ($\tau = RC$).

c. Increase Spacing Between Wires

- i. Greater spacing reduces fringing and coupling capacitance, which directly decreases delay ($\tau = RC$).

d. Reduce Junction Capacitance

- i. Minimize source/drain diffusion area to lower parasitic capacitance.
- ii. Use shallower diffusion regions to reduce (C_d) (diffusion capacitance).

e. Use Lower-K Dielectrics

- i. Reducing the dielectric constant (k) lowers parasitic capacitance (C) in the RC model, decreasing delay.

f. Optimize Gate and Diffusion Capacitance

- i. Properly size transistors to balance gate capacitance (C_g) and diffusion capacitance (C_d).
- ii. Avoid excessive transistor width to reduce unnecessary capacitance.

g. Reduce Resistance with Wider Wires

- i. While narrow wires reduce capacitance, they increase resistance.
 - ii. Balancing wire width optimally reduces RC delay.

- h. Use Multi-Fingered Transistors
 - i. Reducing gate length (L) while increasing number of fingers keeps effective width constant while lowering capacitance.

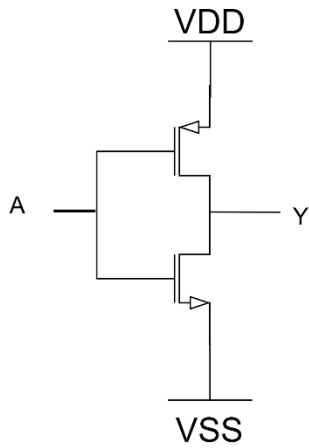
- i. Apply Shielding Techniques
 - i. Grounded shields between signal lines can reduce coupling capacitance.
 - ii. Shields also help in controlling crosstalk and noise.

2. Technique to Fit the Design Within the Cell Template

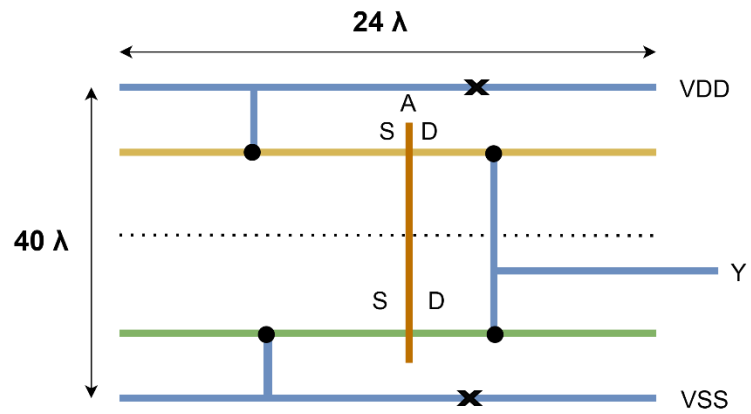
a. When a transistor in a layout exceeds the allowed height limit of the cell template, transistor folding (also called multi-fingered transistors) is used to increase the width while maintaining the height within the constraint. Instead of using a single large transistor that does not fit in the given height, the transistor is split into multiple narrower transistors (fingers) arranged side by side. This technique keeps the height within the allowed limit while increasing the overall width. The transistors are connected in parallel to maintain the same effective width and electrical characteristics as a single wide transistor.

Example: A 480 nm wide PMOS transistor can be folded into two 240 nm wide transistors placed side by side

b.



Before Folding



After Folding

